

NPTEL Course on "Computer System Optimization for ML"

Lecture #32

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Agenda

- Recap
- Chunked Pre-allocation (Sarathi)

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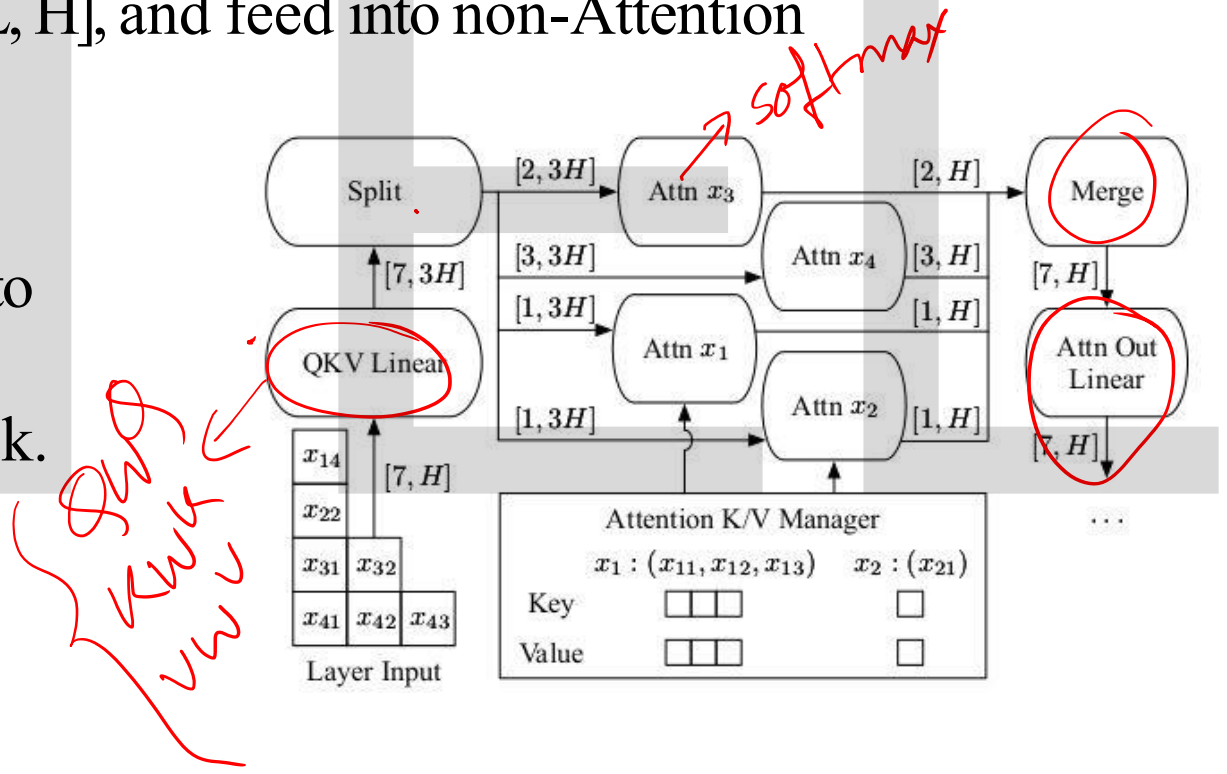
Solution: Selective Batching

Instead of batching **all** tensor operations, only batch the ones you can.

Flatten input tensors into 2-D tensor of shape [∑L, H], and feed into non-Attention operations (Linear, LayerNorm, Add, GeLU).

For Attention operations, split flattened tensor into individual requests and run Attention operation separately. Once completed, merge requests back.

Attention K/V manager provides Keys/Values separately for each request.



Stall free scheduling without compromising on throughput?

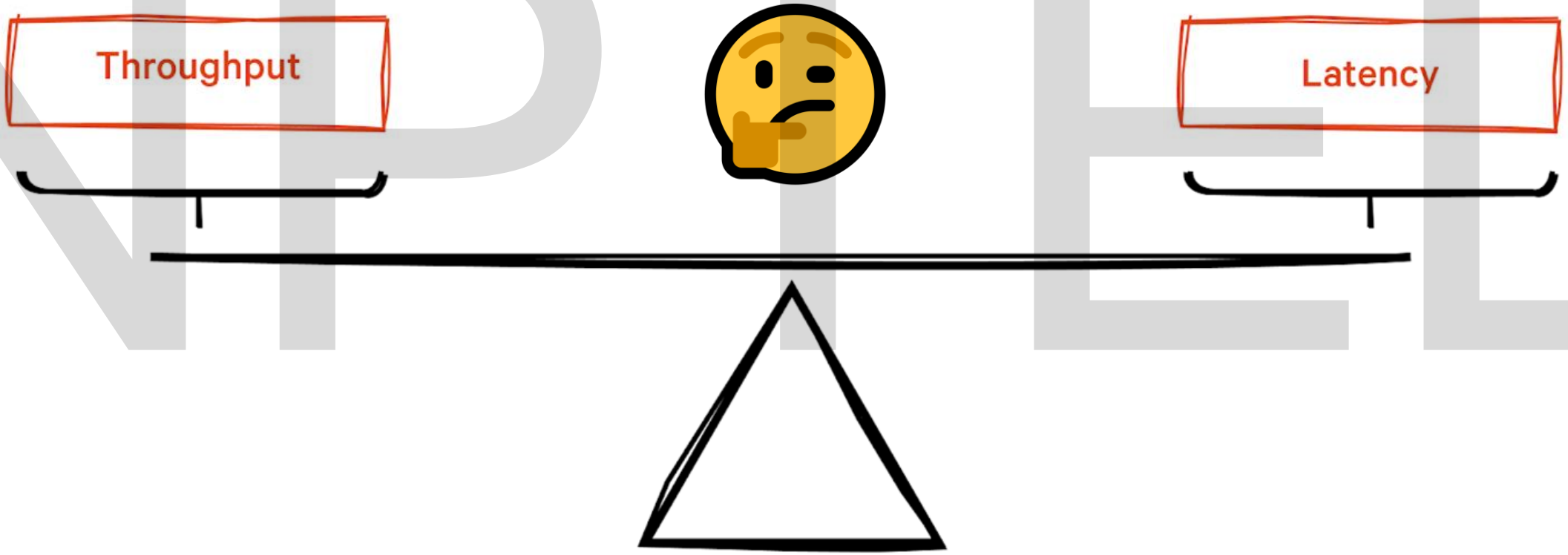


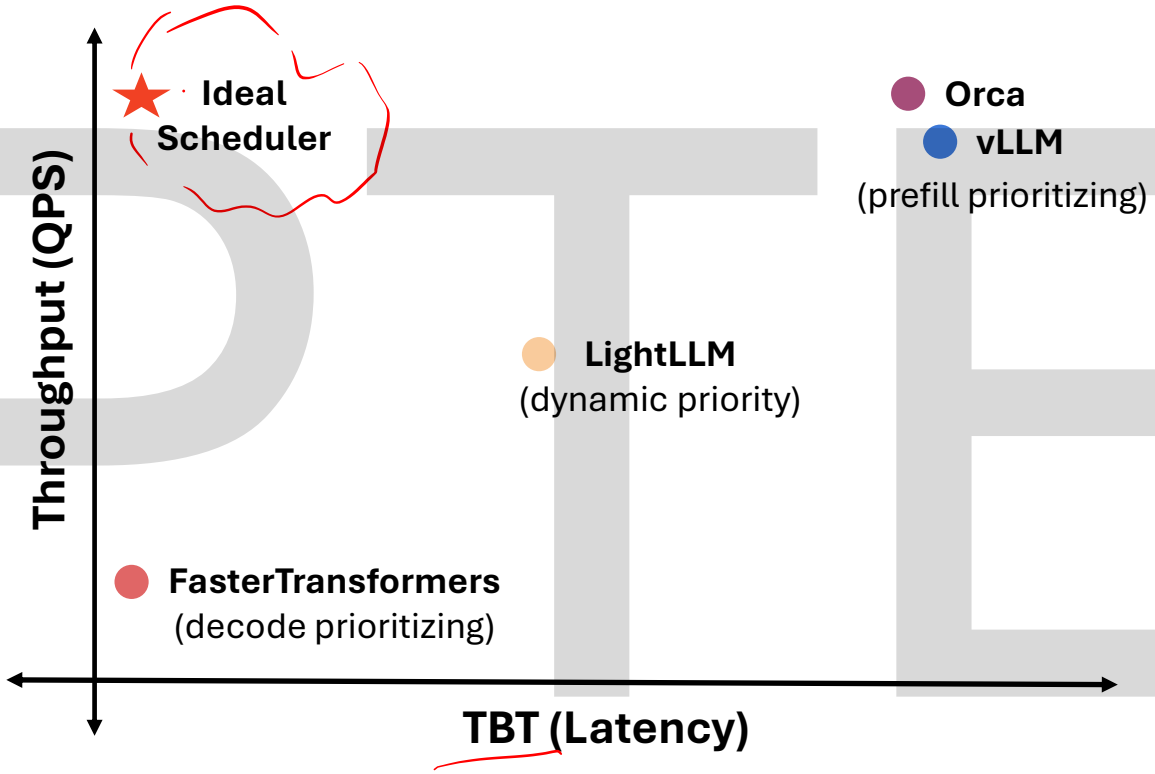
Image credits: Redpanda

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- **Chunked Pre-allocation (Sarathi)**

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Scheduling trade-off

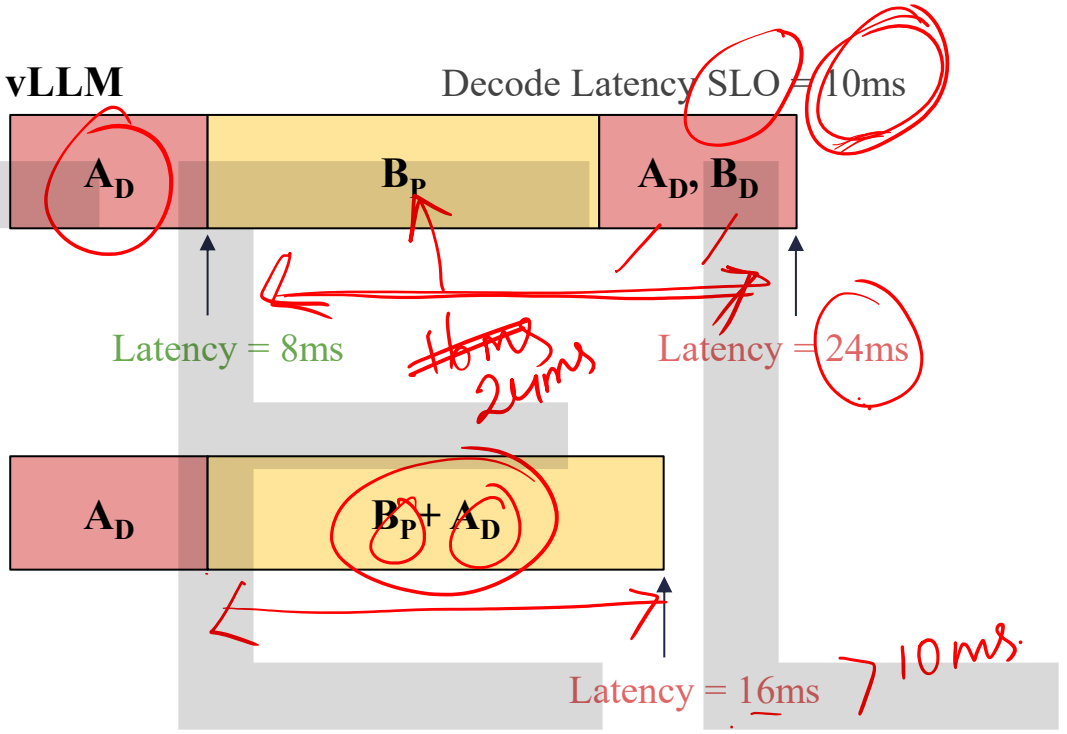


Existing systems compromise on either throughput or latency

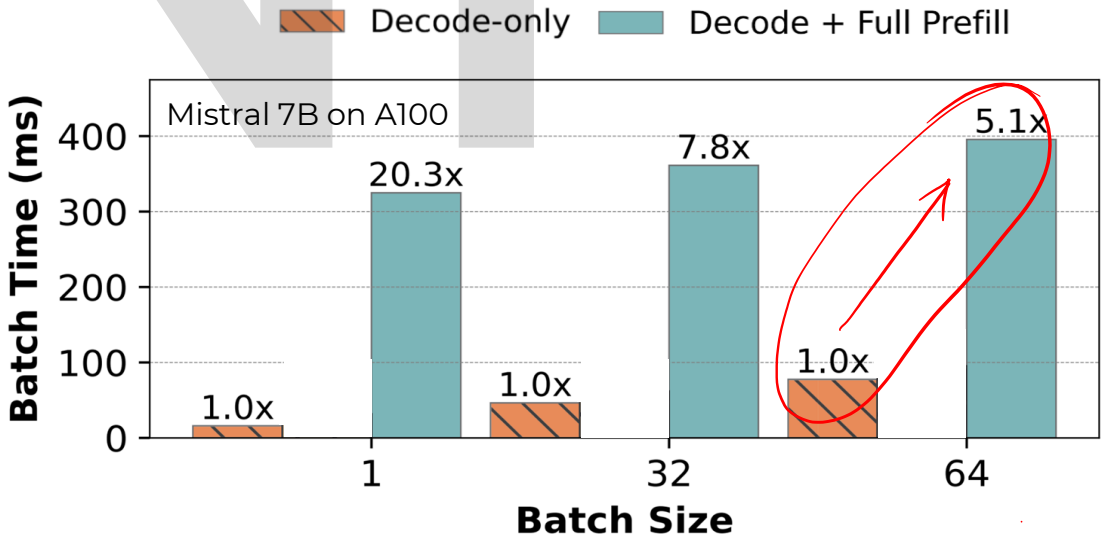
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Mixed Batching

Combine the prefill and decode computation



But, we still violate SLOs – **high latency** due to combining long prefills with decodes

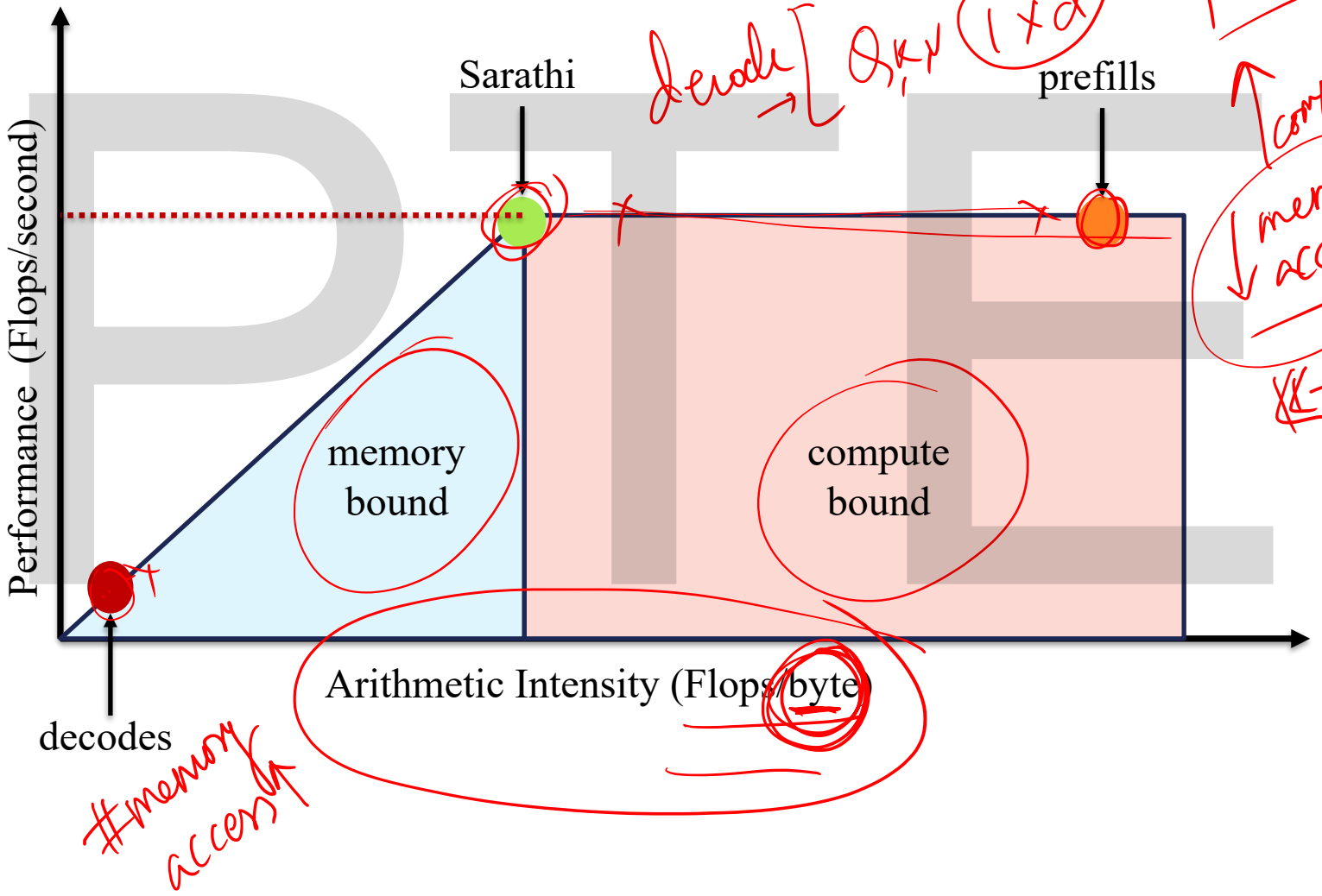


Key Insight

Chunked prefills : Prefills can be split into smaller chunks that marginally add to the decode batch time

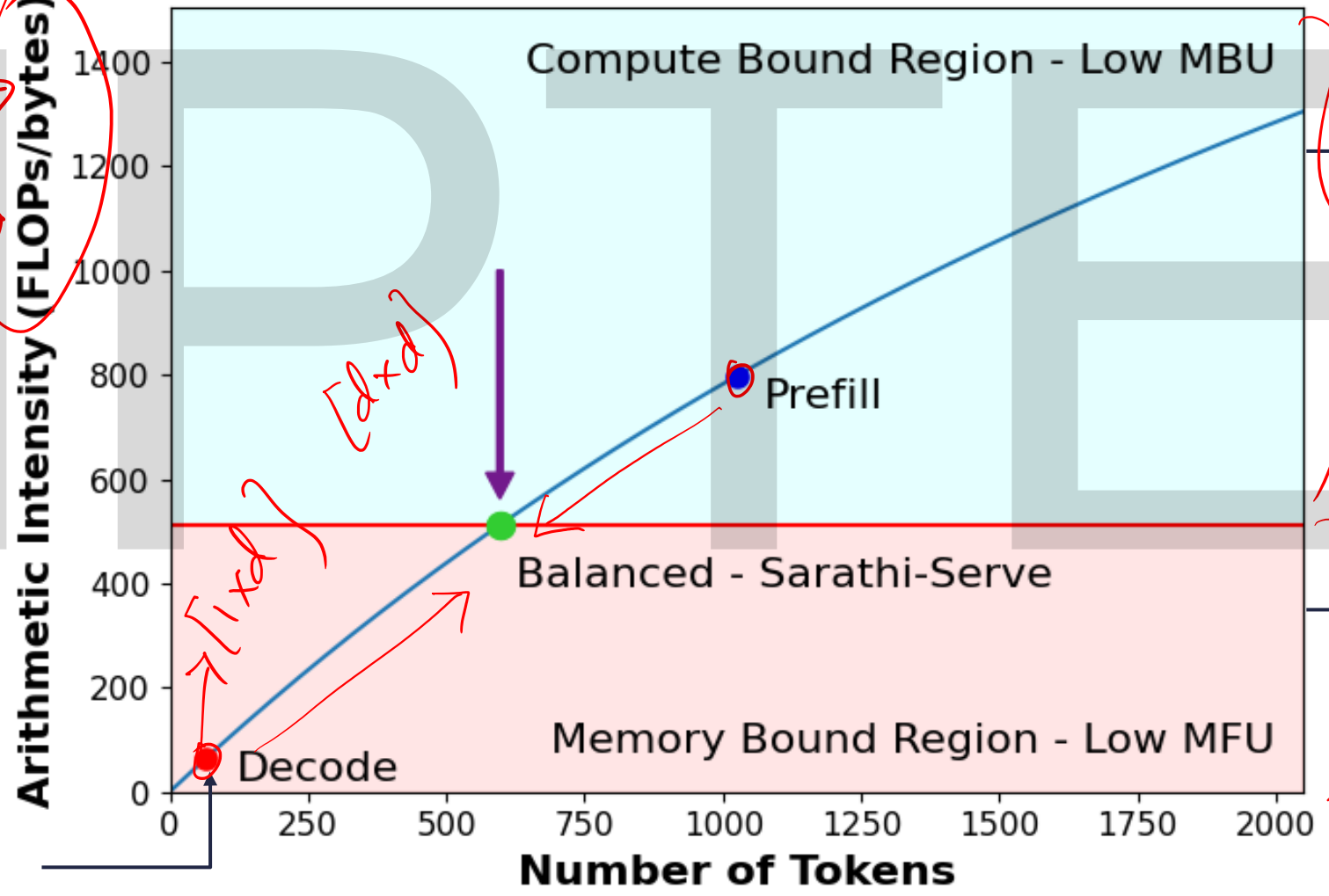
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Exploit arithmetic intensity slack



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🔎 Observation: Arithmetic Intensity Slack



Latency dominated by compute

Grows linearly with num tokens

Latency dominated by weight fetch time

Independent of num tokens

Constrained due to memory overhead in decode phase



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)